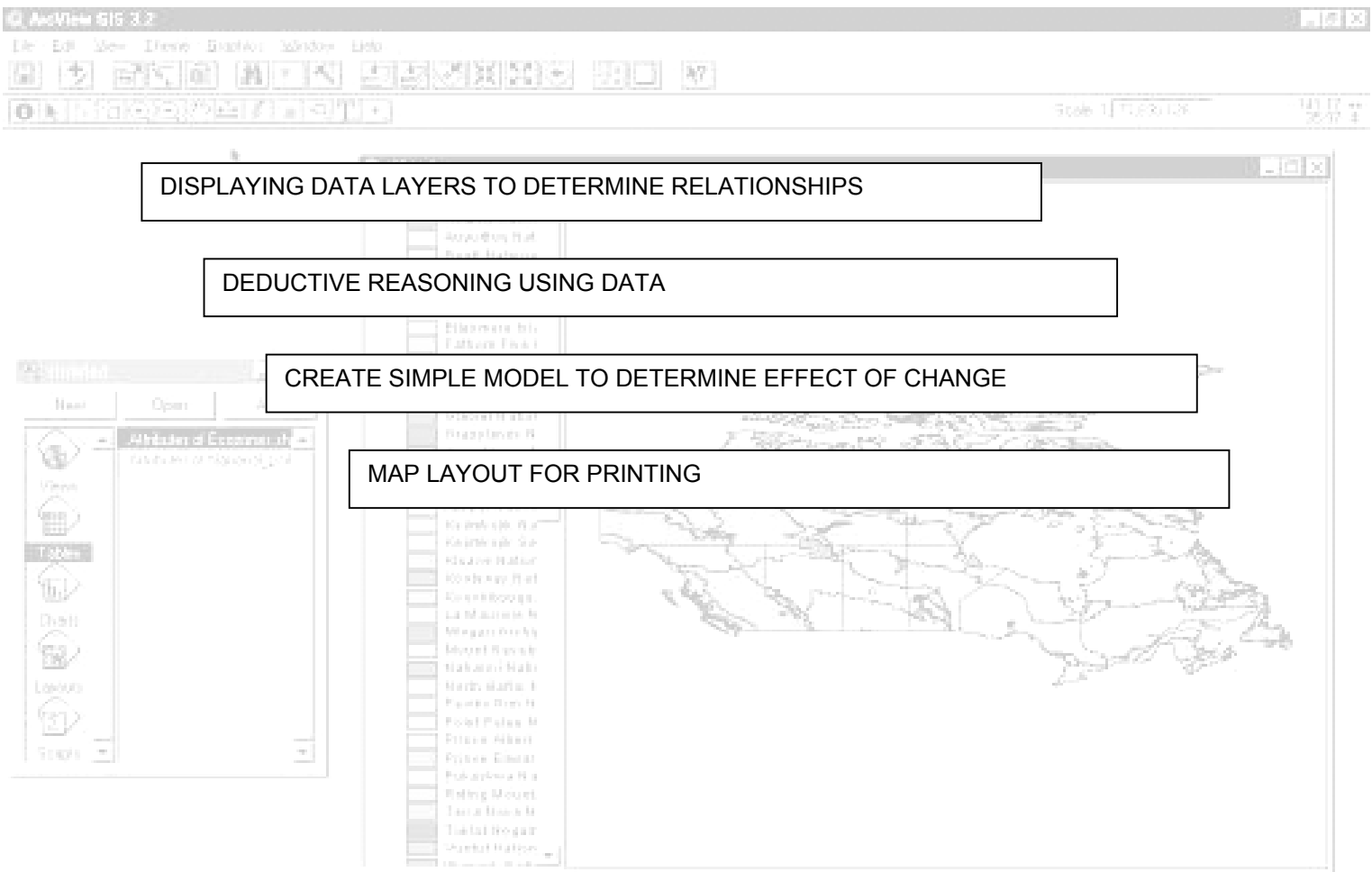


Climate and Precipitation



GIS Information

Here's a list and explanation of important features in ArcView:

Project : In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

View  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

Tables  : Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

Layout  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View – Tools

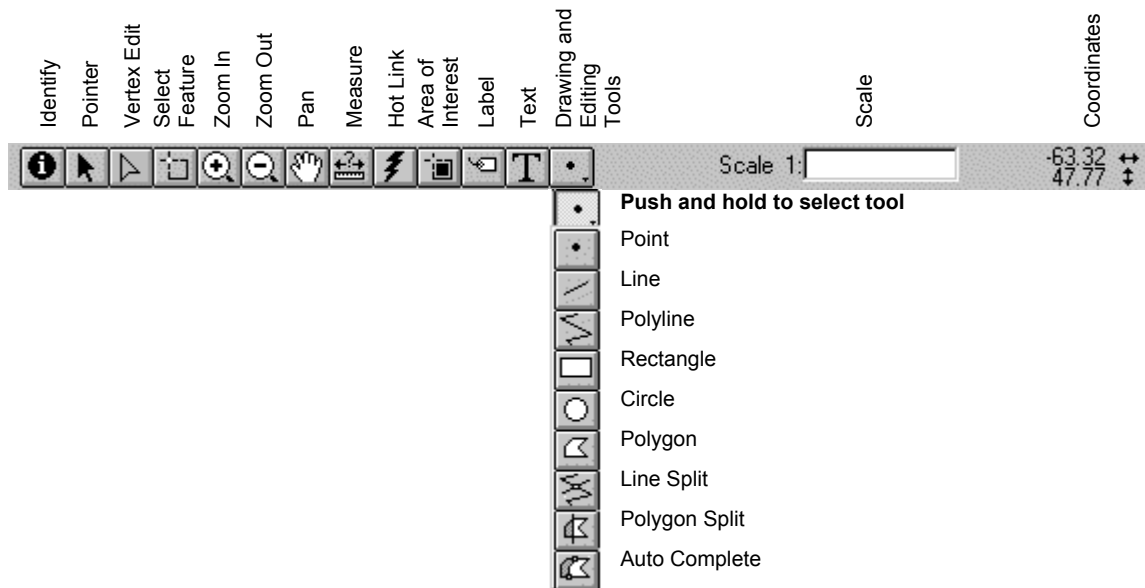


Table - Buttons



Table - Tools

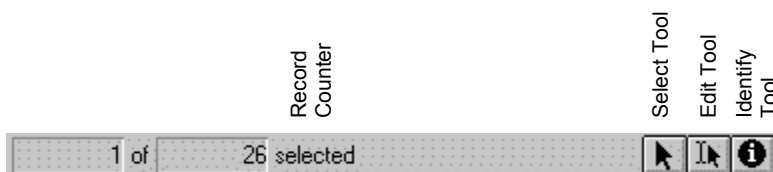


Chart - Buttons

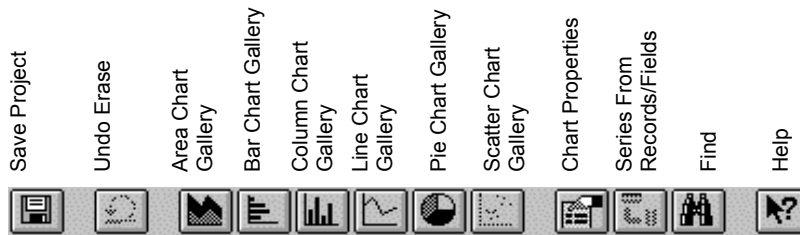
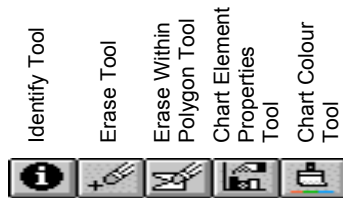
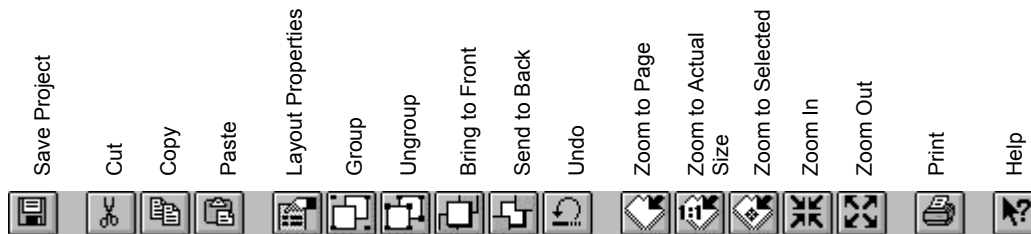


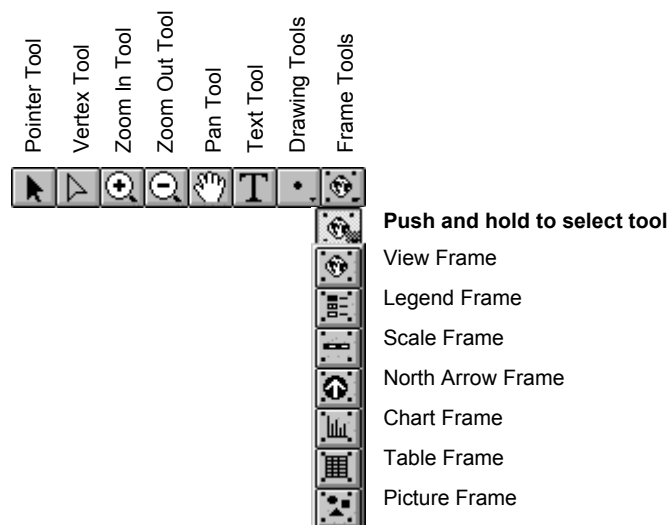
Chart - Tools



Layout - Buttons



Layout - Tools



For more Information: <http://www.esricanada.com>

Getting Started

Before you begin, the lesson data folder must be **copied** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

! (The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).

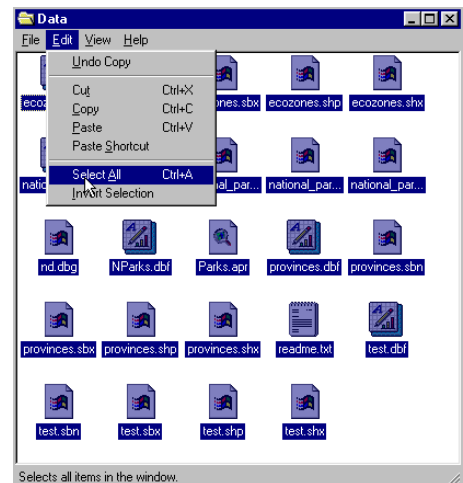
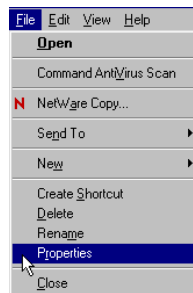
To do this, use the file manager program on your computer. For Windows-based systems this will be **My Computer / Windows Explorer /**.

Once the data folder has been **copied** to the required destination the files must be made usable by removing the **read only** attribute. For Windows-based systems follow these instructions:

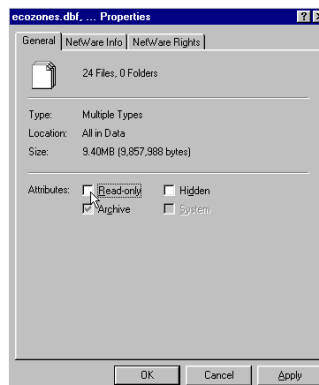
! (The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)

1. Open your new data folder.
2. Choose **Edit > Select all** (they will all appear Blue)

3. From the menu choose:
File > Properties



4. Remove the check mark in the box beside the label **read only**.



5. Click **Apply**, then **OK**.

The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

! Denotes an observation.

? Denotes a question to be answered on the Student Activity Sheet.

STARTING THE PROJECT

This lesson will emphasize using geomatic data to determine relationships between data, form conclusions through deductive thinking, create a simple model and determine its effects. Let's begin by looking at the total precipitation by ecozone.

STARTING ArcView

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*
2. Click **O.K.**

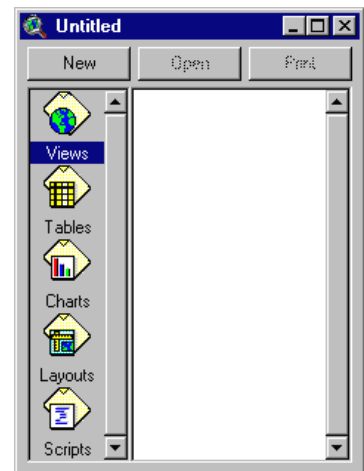


When the program starts, it automatically opens a project window titled "**Untitled.**"

If you like, you can maximize the **ArcView** window to fill the screen.

! In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.

3. Now you are ready to add documents to your new project.
4. With the "**Views**" icon selected, click on **New** and maximize the **View1** window.



5. Click the **Add Theme**  button and navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

Admin.shp
Zones_climate.shp

6. Click **O.K.**

This will add these two themes to the table of contents of the new view. They are now available for display and editing.



7. Click and drag to move the themes up or down in the table of content to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line and point themes should be on top of polygon files to be visible.

8. Click on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be change by clicking on other themes.

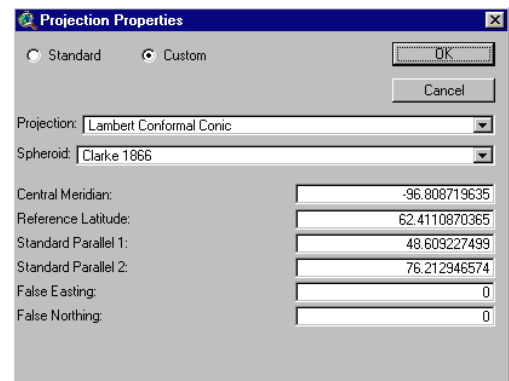
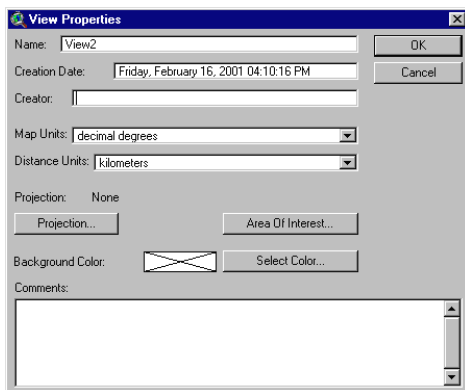
9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project**. Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **.apr** means that you are saving your work as a project that can be opened up in the future).

ArcView - Changing the Map Projection

! A **Map Projection** defines the relationship between the map coordinates and the geographic coordinates, (latitude and longitude). Our current display shows what a map of Canada looks like with only a geographic projection.

Now we will give our map “projection coordinates” to make it look more like the maps which are familiar to us.

1. From the **View** menu, choose **Properties**.
2. Select **Properties**.
3. Select **Map Units** list. Change the **map units to decimal degrees**.
4. Select **Distance Units** list. Change the **distance units to kilometers**
5. Click the **Projection** button.
6. Select **Custom**.
7. Select **Lambert Conformal Conic** from the projection list.



! Take a look at the parameters for this projection. Parameters are the descriptions of how the projection was made.

Note that a spheroid was used. (The earth is a slightly flattened sphere toward the poles). **Conformal** implies that all angles are indicated correctly. All angles measured on the earth's surface are given the same values on the map.

Conical means in the shape of a cone. In this projection, the relative positions on a globe are transferred to a plane in the shape of a cone around the globe. See how the longitudinal lines are closer together at the top of the image, that is, they are not parallel.

The **central meridian**, **false easting** and **false northing** are used to define the X (easting) and Y (northing) coordinates of the map. The **reference latitude** and **standard parallel 1 and 2** are used to determine the origin of the projection (i.e. the point from which you are viewing the globe).

8. Click **O.K.**

The bottom of the **View Properties** window should look like this:



9. Click **O.K.**

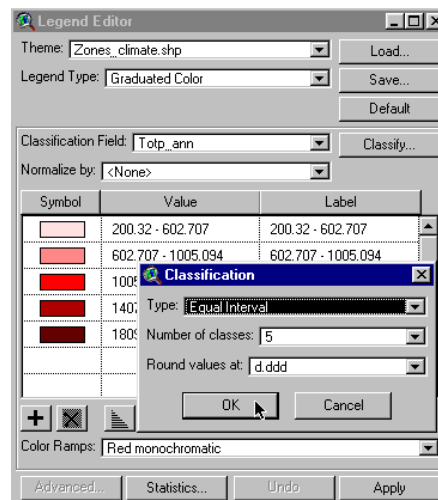
! Note the difference in the map presentation.

? How is it different from the non-referenced map?

ArcView - DISPLAYING DATA LAYERS TO DETERMINE RELATIONSHIPS

First let's make a map to show total annual precipitation by ecozone.

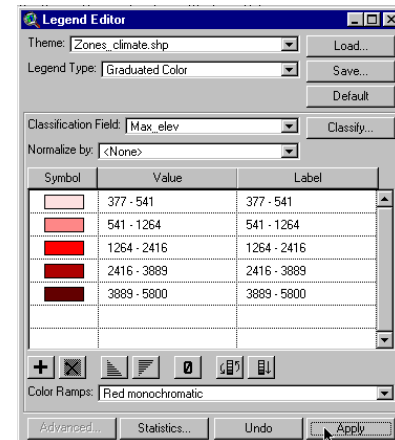
1. Double click **zones_climate** theme.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.
5. Select **totp_ann**.
6. Click **Classify**.
7. Open **Type** list.
8. Select **Equal Interval**. **O.K.**
9. Click **Apply**.
10. Close **Legend Editor** window.



- ? Can you identify the ecozone with the highest annual precipitation?
- ? Which has the second highest?
- ? What do these two wettest ecozones have in common?
- ? Why does Newfoundland have lower precipitation than the rest of the Atlantic Provinces?

! Orographic Precipitation is also known as relief precipitation. It is precipitation caused by rapid condensation in moist air that cools as it rises up the side of mountains. Let's now investigate orographic precipitation in Canada.

1. Double click **zones_climate** theme.
2. Select **Graduated Colour**.
3. Open **Classification** list.
4. Select **max_elev**.
5. Click **Apply**.
6. Close **Legend Editor** window.



- ? Which ecozones do you suspect have orographic precipitation?

Compare the ecozones, which have the highest precipitation with the ones that have the highest elevation.

- ? Which ecozone has the greatest orographic precipitation?
- ? What other characteristics might contribute to orographic precipitation?

! As air sinks down the leeward side of a mountain barrier it becomes warmer. As it warms, it can hold more moisture. As well, the air is dry from the orographic precipitation effect on the air as it rose over the mountain barrier. This low precipitation region on the leeward side of the mountains is called a **rain shadow**.

- ? From the total precipitation map can you identify two ecozones that have rain shadow components?

An important element of climate is temperature. Let's look at temperature by ecozone.

Following the same procedure as before,

1. Double click **zones_climate** theme.
2. Select **Graduated Colour**.
3. Open **Classification Field** list.
4. Select **Tmean_an**. (Mean annual temperature)
5. Click **Apply**.
6. Close **Legend Editor** window.

- ? Can you describe the pattern of mean annual temperature across Canada?

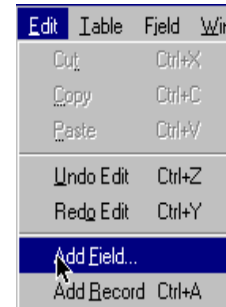
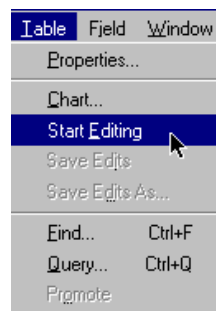
? Can you describe the maritime and continental effects?

DEDUCTIVE REASONING USING DATA

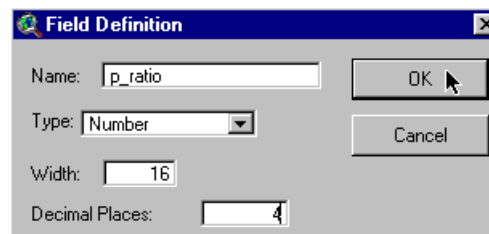
Precipitation is made up mostly of either rain or snow. Let's look at the ratio of snow over rain as a measure of temperature.

1. With the **zones_climate** theme active (raised),
2. click **Open Theme Table**  button.

3. From the **MENU** item, select **Tables**.
4. Click **Start Editing**.
5. Select **MENU** item **Edit**.
6. Click **Add Field**.

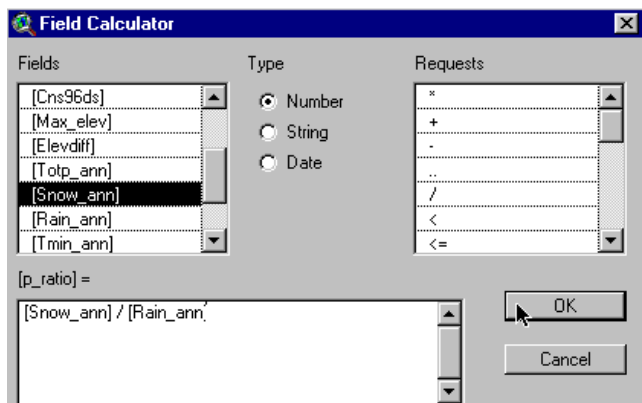


7. Enter name **p_ratio**. (Precipitation ratio)
8. Enter **Decimal Places 4**.
9. Click **O.K.**

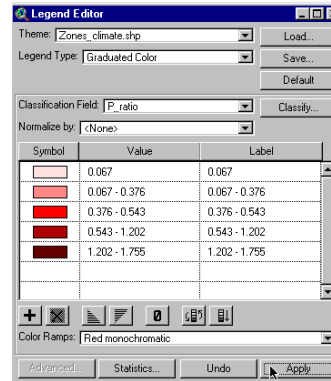


10. Click **Calculate**  button.

11. **Double click snow_ann.**
12. **Double click /.**
13. **Double click rain_ann.**
14. Expression should read **[snow_ann]/[rain_ann]**
15. Click **O.K.**



16. Select **MENU** item **Table**.
17. Click **Stop Editing**. To **Save Edits** click **yes**.
18. Close table.
19. **Double click** *zones_climate* theme.
20. Select **Graduated Colour**.
21. Open **Classification Field** list.
22. Select *p_ratio*.
23. Click **Apply**.
24. Close **Legend Editor** window.



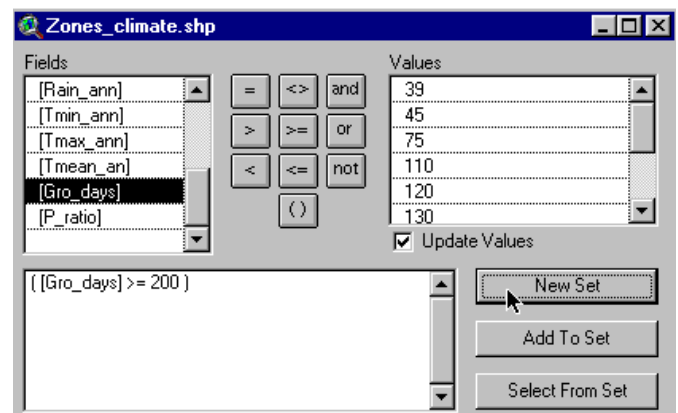
- ? Can you describe the distribution of the snow/rain ratio?
- ? What does a high ratio mean?
- ? How does this relate to temperature?
- ? What about a low ratio?

CREATION OF SIMPLE MODEL TO DETERMINE EFFECT OF CHANGE

Climate affects us in many ways. One of the most important is in agriculture and the food we grow. Let's look at the relationship between agriculture and climate.

Crop X needs at least a 200-day growing season. By looking at the data for number of growing days in each ecozone, we can determine where crop X can grow assuming the soil and other conditions are favorable.

1. Click **Query Builder**  button.
2. **Double click** *gro_days*.
3. Click the **greater than or equal (>=)** button.
4. Type in **200**.
5. Expression should read **[gro_days]>= 200**.
6. Click **New Set**.
7. Close.

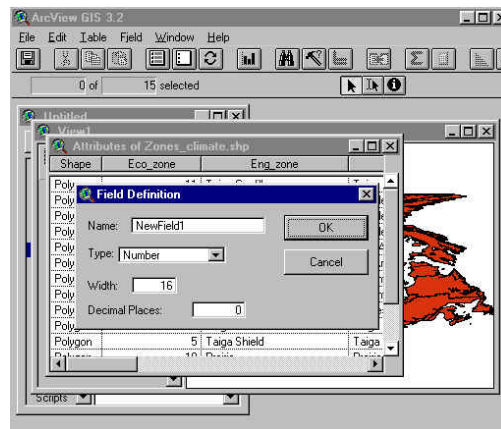


? Identify the ecozones where crop X could be grown if it requires more than 22 growing days?

A climate issue that is of great concern is ***climate change***. This refers to the long-term (over decades) change in our climate caused by global warming.

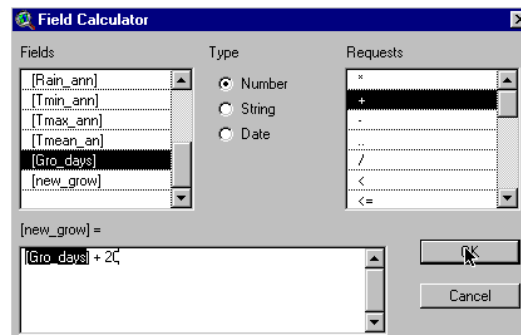
Assume that global warming will lengthen the growing season by 20 days. Let's compute a new growing season.

1. Open **Theme** table.
2. Select **MENU** item **Table**.
3. Click **Start Editing**.



4. Select **MENU** item **Edit**.
5. Click **Add Field**.
6. Enter name **new_gro**. Click **O.K.**

7. Click **Calculate**  button.
8. **Double click** **gro_days**.
9. **Double click** **+**.
10. Type in **20**.
11. Expression should read **[gro_days] + 20**.
12. Click **O.K.**
13. Select **MENU** item **Table**.
14. Click **Stop Editing**. To **Save edit** click **Yes**.
15. Close table.



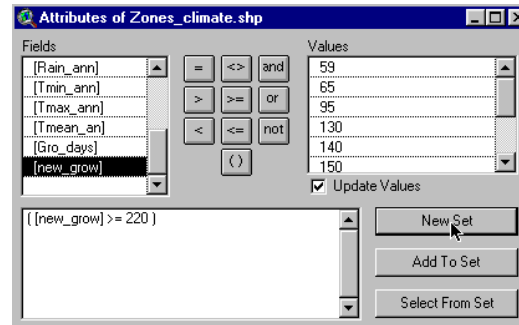
16. Click **Query Builder**  button.

17. Double click **new_gro**.

18. Click **>=** button.

19. Type in **220**.

20. Click **New Set**. Close.



? With the addition of 20 more growing days as a result of climate change, how many ecozones can now support growing Crop X?

? Identify the ecozones, which have been added by climate change.

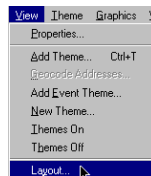
ArcView - MAP LAYOUT FOR PRINTING

Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

SET UP MAP

1. Click Menu **View**.

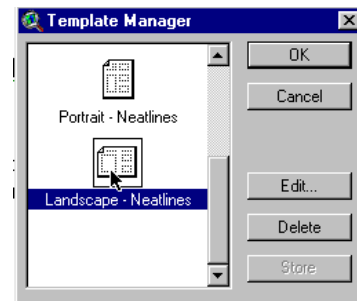
2. Click **Layout**.



3. Select the **Layout Template** you want to use (One of the landscape templates is suggested) .

4. Click **O.K.**

5. Maximize the Layout screen.



EDIT TITLE

1. Double click on the title **View1**.

2. Edit the title to reflect the purpose of the map, e.g., forestry, and population.

3. Click **O.K.**

4. Resize the title using the corner handles and move it into a position that is visually balanced.

5. Click and drag to reposition the legend if necessary.

EDIT NORTH ARROW

1. Double click on the **north arrow** and select a different presentation. Ensure that the arrow is pointing in the representative north direction of the map.

2. Click and drag to reposition or rotate (using rotation function) as needed.

EDIT SCALE BAR

1. Double click the *scale bar*.
2. Choose a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. Change the units to **kms**.
4. Find a combination of style, scale and divisions that is visually pleasing.
5. Click and drag to position to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. Click on the **Text** tool.
2. Click at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., "produced by (your name, date)".
4. Select desired formatting information such as **alignment**, **spacing** and **rotation**.
5. Click **O.K.** when it appears, as you want it.
6. Click on **Pointer** tool.
7. Select the new text
8. Click once to change location or size. Click twice to change content.

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

You only have to do this if you want to save your project on the computer or on a disk.

1. Click menu **Layout**.
2. Select **Properties**.
3. Change the "name". Click **O.K.**

ARRANGING YOUR LAYOUT

! Experiment with the layout buttons (use the tools page provided at the beginning of this lesson if you are unsure of their function).

- to **group** (tie pieces together so they move as one)
- to **ungroup** (break apart the ties so that individual pieces move)
- **bring to front** (change order that pieces were laid down, e.g., lines on top of polygons)
- **send to back** (change order that pieces were laid down, e.g., polygons behind lines)
- **zoom** (make image bigger or smaller)
- **neat line** (border around layout)

PRINT MAP LAYOUT

1. Click **Print** button.
2. Click **Setup**.
3. Select your printer properties as outlined by your teacher.
4. Click **O.K.** to print.

! **Note** that the map can also be saved to a printer file.